PRODUCT USER MANUAL

For Global Ocean-Delayed Mode in-situ Observations of surface (drifters and HFR), surface and subsurface (Argo drift) and sub-surface (vessel-mounted ADCPs, gliders) water velocity

INSITU_GLO_PHY_UV_DISCRETE_MY_013_044
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Contributors : H. Etienne, Christine Boone, Patricia Zunino, N. Verbrugge, E. Reyes, P. Rotllán, A. Rubio, L. Solabarrieta, L. Corgnati, C. Mantovani, J. Mader, Thierry Carval, Corentin Guyot

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MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

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3.2	16/02/2022	§II.2	Update frequency added	Nathalie Verbrugge	Stéphane Tarot
3.3	23/05/2022	All	product and datasets renaming + new template	N. Verbrugge, A. Rubio	S. Tarot
3.4	06/2023	All	Add Argo dataset + table numeration update + minor corrections	N. Verbrugge	S. Tarot

3.5	05/2024	All	New upstream data for drifters New NetCDF format New gliders dataset	N. Verbrugge, P. Zunino, C. Guyot, L. Solabarietta	S. Tarot
3.6	05/2025	All	Add new variables in the drifter dataset: EWCT_UNCERTAINTY, NSCT_UNCERTAINTY, DROGUE_TEST, and change in the definition of EWCT_QC and NSCT_QC	P. Zunino, N. Verbrugge, L. Solabarrieta	S. Tarot

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GLOSSARY AND ABBREVIATIONS

TAC	Thematic Assembly Center
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
HFR	High Frequency Radar
ADCP	Acoustic Doppler Current Profiler
GDAC	Global Data Assembly Center
EGO	Everyone's Gliding Observatories

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This Product User Manual describes the INSITU_GLO_PHY_UV_DISCRETE_MY_013_044 product distributed by the Copernicus Marine Service In Situ Thematic Assembly Centre (In Situ TAC). This manual describes how it is built, what is the content, what data services are available to access them, and how to use the files.

The *In Situ Global ocean* delayed mode product designed for reanalysis purposes integrates the best available version of **in situ data** for **ocean surface currents and current vertical profiles**.

This product concerns **six delayed time datasets** dedicated to near-surface currents measurements coming from two platforms (Lagrangian surface drifters and High Frequency radars), velocity profiles within the water column coming from the Acoustic Doppler Current Profiler (ADCP) platform and gliders, and velocity at surface and around 1000m (parking depth) coming from Argo floats trajectories:

cmems_obs-ins_glo_phy-cur_my_drifter_PT1H, named drifter in the rest of the document: delayed time zonal and meridional raw velocities measured by drifting buoys, quality flags and metadada. These surface observations are collected by Coriolis (the French Data Assembly Center), and velocities are estimated by the In Situ TAC (CLS) using the Elipot et al. (2016) methodology. In addition, a wind slippage correction, from 1993 until the end of the time series, is provided by the In Situ TAC (CLS)

cmems_obs-ins_glo_phy-cur_my_radar-radial_irr, named radar radial in the rest of the document: near-surface radial velocities of High Frequency Radars (HFR), quality flags and metadata. These surface observations are part of the European HF radar Network (see Mader et al, 2017 and Corgnati et al., 2018).

cmems_obs-ins_glo_phy-cur_my_radar-total_irr, named radar total in the rest of the document: near-surface zonal and meridional components of HFR derived total velocities, quality flags and metadata. These surface observations are part of the European HF radar Network (see Mader et al, 2017 and Corgnati et al., 2018) and the HFR US networks.

cmems_obs-ins_glo_phy-cur_my_adcp_irr, named adcp in the rest of the document: three-dimensional water current velocities over a depth range along a vessel underway trajectory or mooring.

cmems_obs-ins_glo_phy-cur_my_argo_irr, named argo in the rest of the document: ocean currents derived from the Andro delayed mode Argo floats trajectories (Ollitraut et al., 2022).

Deep current is calculated from floats drift at parking depth, surface current is calculated from float surface drift.

cmems_obs-ins_glo_phy-cur_my_glider_irr, named glider in the rest of the document: marine currents derived from glider trajectories in the EGO (Everyone's gliding observatories) GDAC dataset. The current is estimated for each glider cycle and its depth is the mean immersion of each cycle.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Table 1 : In Situ TAC UV MY product and datasets

Product code	**INSITU_GLO_PHY_UV_DISCRETE_MY_013_044** For Global Ocean- Delayed Mode in-situ Observations of surface (drifters, Argo drift and High Frequency radars) and sub-surface (vessel-mounted ADCPs, glider drift and Argo drift) water velocity					
Dataset[1]	**drifter**	**radar_radial**	**radar_total**	**adcp**	**Argo**	**glider**
Spatial coverage	*global*	*European Seas from coast to up >200 km (depending on the operating frequency)*	*European and US Seas from coast to up >200 km (depending on the operating frequency)*	*global*	*global*	*global*
Spatial resolution	*discrete*	*Gridded (Typically ranges from a few hundred meters to 5-6 km, depending on HF Radar operating central frequency and bandwidth)*		*discrete*	*discrete*	*discrete*
Temporal resolution	*hourly*	*Hourly (exceptions with 15' or 30')*	*Hourly (exceptions with 15' or 30')*	*2 minutes*	*Typical resolution of 10 days*	*It depends on the glider deployment configuration. Several measurements per day*

[1] File type is detailed when applicable

Temporal coverage	*12/1979 to 6 months to 1 year from present*	*01/2009 to 6 months to 1 year from present (but not for all the systems)*	*01/2009 to 6 months to 1 year from present (but not for all the systems)*	*04/01/2001 to 1 year to 2 years from present*	*Jan 2000 to 6 to 12 months before present*	*06/2004 to 6 to 12 months before present*
Number of vertical levels for UV	*1*	*1*	*1*	*Variable* *Several depths up to 200m for 150kHz ADCP; up to 850m for 38kHz ADCP*	*2 different depths: surface and parking depth*	*1 (mean depth of the profile)*
Variables	*Zonal and Meridional Velocities at 15-m depth or surface, Zonal and Meridional wind-slippage correction. QC variables and metadata (global attributes)*	*Zonal and Meridional components and magnitude and direction of radial (referred to the individual measuring HFR stations) velocities at the surface (actual depth depending on the operating frequency), standard deviation of zonal and meridional components of the radial velocities at the surface. QC variables and metadata (global attributes)*	*Zonal and Meridional Velocities at the surface (actual depth depending on the operating frequency), standard deviation of zonal and meridional velocities at the surface Geometrical Dilution of Precision (GDOP). QC variables and metadata (global attributes)*	*Vertical profiles of eastward_sea_water_velocity, northward_sea_water_velocity, upward_sea_water_velocity*	*Zonal and Meridional Velocities at surface and sub-surface derived from Argo trajectory files with QC variables and metadata.*	*Zonal and Meridional Velocities at sub-surface derived from Glider trajectory files with QC variables and metadata.*
Product type	*Observations, delayed-mode, multi-year*					

Update Frequency	*The product is updated twice a year: in June for a temporal extension; in December for a full reprocessing or/and a temporal extension. A full reprocessing is associated to different improvements (qualification procedures, new data, new methodology,…)*
Delivery time	*biannual*
File Format (see sect. 3))	*NetCDF4 classic model*
Delivery mechanism	*Copernicus Marine Service Information Service (see sect. 4))*

b) Production System Description

a. Drifters trajectories

The Delayed Time Surface Drifters Dataset is constructed from historical drifter data of the Copernicus Marine Service product INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030 / cmems_obs-ins_glo_phybgcwav_mynrt_na_irr/ history/DB/. This product is yielded by Coriolis-Ifremer who operates as Data Assembly Center (DAC). They collect the drifter data, verify that the drifters are operational, they assemble and quality control the data and make them available. This product includes historical drifter data which locations were quality controlled, including visual inspection, in the framework of the C-RAID project (Zunino et al., 2024).

Satellite-tracked surface drifting buoy locations are used for the calculation of the velocities. They are first edited, which means to remove spurious positions following automatic tests (Lumpkin et Pazos; 2006; Lumpkin et al., 2012). Then, positions are time - interpolated every hour and velocities derived, following two methodologies, which depend on the drifter location transmitter system. On the one hand, the Weighted Maximum Likehood Estimates (WMLE) for drifters with Argos transmitter system. On the other hand, a 2-D implementation of a robust regression method called Locally Weighted Scatterplot Smoother (LOWESS) for drifters with quality-controlled GPS locations. Both methodologies are described in Elipot et al., 2016.

Wind slip correction is then computed by CLS (Rio, 2012), added to the database as well as quality control flags, before the final dissemination of the data to Copernicus Marine Service In Situ TAC GLO in standard file format.

b. High-Frequency Radars surface currents

The Delayed-Mode Surface currents from HF radars derive from the historical data collected, analysed and pre-processed by AZTI and the European HFR Node, before the final dissemination of the data to Copernicus Marine Service In Situ TAC GLO in standard file format. The European HFR node delivers in near real-time and on an hourly basis, maps of surface current velocities from the HF radars that are actively processing and/or delivering their (formatted or raw) data to the node. The European HFR node also collects and processes near real time HFR data for advanced QA/QC and aggregation of files to build the REP dataset of historical surface current velocity data from those operators connected to the node. In the European framework, the European HFR Node is now managing data from 21 HFR EU networks (built by 64 radar sites) and 5 HFR US networks (just totals; not radials). Of these 26 networks, 18 EU networks are sending radial and total data in NRT, 1 EU system stopped in 2016 and just provided historical data, and all US networks (5) and 2 EU network are sending NRT total data. Moreover, 12 EU networks have provided time series of historical data before 2019, the beginning year of the NRT dataflow.

HFRs provide surface ocean current data only relative to the surface within an integration depth ranging from tens of centimeters to a few meters, depending on the operating central frequency and averaged over a time interval (mainly around the cardinal hour) and delivered at hourly basis.

Radial velocities are measured on a polar grid and then remapped on a Cartesian grid. The final product is a map of the zonal and meridional components of the radial ocean currents on a regular grid in the area covered by the individual radar stations.

Total velocities are derived by least square fitting to the radial velocities, (measured by at least 2 individual stations), onto a Cartesian grid. The final product is a map of the zonal and meridional components of the ocean currents on a regular grid in the area of overlap of two or more radar stations.

c. *Acoustic Doppler Current Profilers velocities*

The reprocessed ADCP product is based on the NRT multiparameter product (INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030) history directory files downloaded from the Global Distribution Unit. Quality flags for the most relevant parameters have been reprocessed using updated quality assessment procedures. This work has been performed for each platform.

Quality control procedures were developed and tested by Ifremer and the European project AtlantOS: Cascade ADCP data delayed mode data processing.

More information on these procedures and on the ADCP product quality assessment is available on

https://www.umr-lops.fr/en/Technology/Software/CASCADE-7.2

d. *Surface and parking depth velocities from Argo trajectories*

The Argo currents dataset is derived from Argo floats trajectories available in Andro dataset (Ollitraut et al., 2022) which is a full reprocessing of Argo floats data. The raw data are the original data received from satellite, decoded and quality controlled into a trajectory file. Each trajectory is checked by a data scientist. These trajectories are more homogeneous than the trajectory files from Argo GDAC (Global Data Assembly Centre), produced by 12 distinct teams (DACs).

This dataset provides currents at the surface and in subsurface, at the parking depth of the floats. The positions used to estimate the velocities are measured during the surface drift (Argos or GPS positioning, no Iridium). A more detailed description of the method is available in the Quality Information Document of the product: INSITU_GLO_PHY_UV_DISCRETE_MY_013_044 (https://doi.org/10.13155/41256).

e. *subsurface velocities from glider trajectories*

The Glider currents dataset is derived from EGO glider trajectory files available in EGO GDAC.

The raw data are the original glider data received via satellite, decoded and quality controlled.

The currents are directly estimated by the glider. For Slocum gliders (Slocum Glider - Woods Hole Oceanographic Institution (whoi.edu)), the best estimated current is selected during the decoding phase by an algorithm developed by Lucas Merckelbach (GitHub - smerckel/gliderflight at 1.2.0), based on the ideal and actual drift times.

A visual inspection of glider currents is performed on the dataset.

This dataset provides subsurface current measurements (one per profile), with the associated depth representing the mean immersion depth.

c) Dataset details

Copernicus In Situ TAC distributes data and metadata following the NetCDF CF convention implementation specified in RD [1] Copernicus Marine In Situ NetCDF Format Manual.

The **parameters** field contains the physical parameters as listed in the RD [2] Copernicus Marine in situ TAC - Physical Parameters List and candidate parameters (not yet approved in the official list).

Information about the global attributes can be found in the RD [3] Copernicus Marine in situ TAC - Attributes list.

a. Content: dimensions

The dimensions are used to represent a real physical dimension (e.g. time, latitude, longitude, depth, etc) but they might also be used to index other quantities (e.g. station)[2]. NetCDF dimensions provide information on the size of the data variables, and additionally the coordinate variables to data. CF convention recommends that if any or all the dimensions of a variable have the interpretations of "date or time" (T), "height or depth" (Z), "latitude" (Y), or "longitude" (X) then those dimensions should appear in the relative order T, Z, Y, X in the variable's definition.

[2] Information provided by https://www.unidata.ucar.edu/software/netcdf/docs/netcdf_data_set_components.html

Table 2- List of the dimensions of the data variables

name	comment	datasets					
		drifter	radar_radial	radar_total	adcp	argo	glider
TIME	Number of time steps.	x	x	x	x	x	x
DEPTH	Number of depth levels. Set to 1 for drifter and HFR data. For drifter equivalent to drogue depth when DROGUE_TEST =1 and equal to 0m when DROGUE_TEST =3, respectively.	x	x	x	x	x	x
LATITUDE	Dimension of the LATITUDE coordinate variable.		x	x			
LONGITUDE	Dimension of the LONGITUDE coordinate variable.		x	x			
STRINGx	Length in characters of the strings used in the data file. It is mandatory that the string length dimension STRINGx has the value of x.	x	x	x	x	x	x
MAXSITE	Maximum number of contributing antennas. Set as an upper bound.			x			
MAXINST	Maximum number of collaborating institutions. Set as an upper bound.		x	x			

REFMAX	Maximum number of external resource linkages. Set as an upper bound.		x	x			
STRLEN	String length for variables containing the cf_role attribute	x			x	x	x

Since HFR data have only one depth layer of measurement, i.e. the surface layer, the dimension DEPTH has size equal to 1 and value equal to 0 meter.

If non-physical variables are present in the data file, e.g. the processing parameters of the HFR device generating the data or the codes of the sites contributing to a total current velocities data, related non-physical dimensions might be defined to expose the variables in the model.

For HFR, more than one STRINGx dimension can be defined, provided that the string length dimension STRINGx has the value of x.

b. Content: variables

Variables are used to store the bulk of the data (an array of values of the same type) in a netCDF dataset.

DRIFTER VARIABLES

This is a global coverage dataset and available variables are listed in Table 3.

Table 3- List of the available variables for the drifter dataset

Variable name	Description	Units
TIME	Date of the data	days since 1950-01-01T00:00:00Z
TRAJECTORY	Trajectory, coordinate variable containing the cf_role	Dimensionless
DEPH	Nominal depth of sea water velocity (indicated in the file)	m
LATITUDE	Latitude of the data position	Degree North
LONGITUDE	Longitude of the data position	Degree East
EWCT	West-East sea water velocity at the drogue depth from the drifter position. When drogue is lost, this is the surface sea water velocity.	$m.s^{-1}$
NSCT	South-North sea water velocity at the drogue depth from the drifter position. When drogue is lost, this is the surface sea water velocity.	$m.s^{-1}$
EWCT_WS	West-East wind slippage correction	$m.s^{-1}$
NSCT_WS	South-North wind slippage correction	$m.s^{-1}$
WS_TYPE_OF_PROCESSING	Wind slippage correction processing method	Dimensionless
EWCT_UNCERTAINTY	Uncertainty of West-East sea water velocity	$m.s^{-1}$
NWCT_UNCERTAINTY	Uncertainty of South-North sea water velocity	$m.s^{-1}$
DROGUE_TEST	(=1: Drogue on; =3: Drogue off ; =0 Drogue lost unknown)	Dimensionless

The total drifter velocity $\mathbf{U}_d$ is decomposed into different contributions:

- geostrophic current $\mathbf{U}_g$
- Ekman/Stokes current $\mathbf{U}_e$ (wind-driven current)
- Remaining ageostrophic current $\mathbf{U}_a$, including wind slip.

When the drogue is lost, the drifter is directly under the influence of the wind stress. Using DUACS geostrophic current from altimetry (https://doi.org/10.48670/moi-00145 products[3]) and an empirical Ekman/Stokes model, a wind slippage correction $\mathbf{W}_s$ is computed since the residual current $\mathbf{U}_r$=($\mathbf{U}_d$- $\mathbf{U}_g$- $\mathbf{U}_e$)$_f$ is expected to be proportional to the wind slippage $\mathbf{U}_s$ (see CMEMS-INS-QUID-013-044 for more details).

This variable is not provided in the Mediterranean Sea nor at the equator for the drogued buoys.

The **WS_TYPE_OF_PROCESSING** is an indication of the way the wind slippage correction is computed which can be selected by the user (values as shown in Table 4):

Table 4 - WS_TYPE_PROCESSING variable's values

Method	Value	Meaning
Optimal	0	If the drifter's measurement is in optimal mode, it is considered that its trajectories is longer than 200 days
Mean	1	The first/last days of the trajectory are completed using the mean value over the buoy time series
Climatology	2	For drifters with trajectories shorter than 60 days a climatological coefficient is used.
Adaptative	3	From drifter's trajectories ranging between 60 and 200 days an adaptative moving correlation window is used.

RADAR_RADIAL & RADAR_TOTAL VARIABLES

This is a dataset covering European and US Seas whose available variables are listed in Table 5 (**mandatory variables**, i.e. variables that are always present in each data file, are listed in bold, recommended variables are listed in plain text).

[3] Geostrophic current only available from 01/01/1993 only

Table 5- gridded radar_radial and radar_total dataset.

Variable name	Description	Units	Dataset	
			radar_radial	radar_total
TIME	**Time of measurement**	days since 1950-01-01T00:00:00Z	**x**	**x**
DEPH	**Depth of the measurement** (from tens of cm to 1-2 m, depending on the operating frequency)	m	**x**	**x**
LATITUDE	**Latitude of the data position**	Degree North	**x**	**x**
LONGITUDE	**Longitude of the data position**	Degree East	**x**	**x**
EWCT	**Surface Eastward Sea Water Velocity** (gridded maps of the surface current velocity component averaged over a time interval, mainly around the cardinal hour).	m/s	**x**	**-**
	Surface Eastward Sea Water Velocity (gridded maps of the zonal component of the radial surface ocean current velocity averaged over a time interval, mainly around the cardinal hour).	m/s	**-**	**x**
NSCT	**Surface Northward Sea Water Velocity** (same as EWCT)	m/s	**x**	**-**
	Surface Northward Sea Water Velocity (gridded maps of the meridional component of radial surface ocean current velocity averaged over a time interval, mainly around the cardinal hour)	m/s	**-**	**x**
EWCS ***(or UACC for phase array systems)***	**Standard Deviation Of Surface Eastward Sea Water Velocity**	**m/s**	**x**	**x**
NSCS (or VACC for phase array systems)	**Standard Deviation Of Surface Northward Sea Water Velocity**	**m/s**	**x**	**x**
CCOV	Covariance of Surface Sea Water Velocity (for	m^2/s^2	x	x

	data measured by Codar systems)			
GDOP	**Geometrical Dilution Of Precision** (QC-related parameter)	**dimensionless**	**x**	**x**
UACC ***(or EWCS for direction finding systems)***	**Accuracy of Surface Eastward Sea Water Velocity**	**m/s**	**x**	**x**
VACC ***(or NSCS for direction finding systems)***	**Accuracy of Surface Northward Sea Water Velocity**	**m/s**	**x**	**x**
NARX	Number of Receive Antennas	dimensionless	x	x
NATX	Number of Transmit Antennas	dimensionless	x	x
SLTR	Receive Antenna Latitudes	Degrees North	x	x
SLNR	Receive Antenna Longitudes	Degrees East	x	x
SLTT	Transmit Antenna Latitudes	Degrees North	x	x
SLNT	Transmit Antenna Longitudes	Degrees East	x	x
SCDR	Receive Antenna Codes	dimensionless	x	x
SCDT	Receive Antenna Codes	dimensionless	x	x
BEAR	**Bearing (away from the instrument) of the measurement**	**Degrees true**	**-**	**x**
RNGE	**Range (away from the instrument) of the measurement**	**km**	**-**	**x**
RDVA	**Radial Sea Water Velocity Away From Instrument** (gridded maps of the surface current velocity radial component averaged over a time interval, mainly around the cardinal hour)	**m/s**	**-**	**x**
DRVA	**Direction of Radial Sea Water Velocity Away From Instrument** (gridded maps of the direction of surface current velocity radial component averaged over a time interval, mainly around the cardinal hour)	**Degrees true**	**-**	**x**
ESPC ***(or HCSS for phase array***	**Radial Standard Deviation of Current Velocity over the Scatter Patch**	**m/s**	**-**	**x**

systems)				
ETMP *(or EACC for phase array systems)*	**Radial Standard Deviation of Current Velocity over the Coverage Period**	**m/s**	**-**	**x**
HCSS *(or ESPC for direction finding systems)*	**Radial Variance of Current Velocity Over Coverage Period**	**m/s**	**-**	**x**
EACC *(or ETMP for direction finding systems)*	**Radial Accuracy of Current Velocity Over Coverage Period**	**m/s**	**-**	**x**

Concerning the variables EWCS, NSCS, UACC and VACC:

- EWCS and NSCS variables are related to the parameters defining the variance of the velocity component measurements present (mainly) in direction-finding HF radar systems (see Rubio et al., 2017). They are computed at each time step, and therefore considered not statistically solid.
- UACC and VACC variables are related to the parameters defining the accuracy of the velocity component measurements and they are mainly present in phased-array HF radar systems. Thus, the two couples of variables are alternative, depending on the system producing the data.

Concerning the variables ESPC, ETMP, HCSS and EACC, it has to be noted that ESPC and ETMP are related to the parameters defining the standard deviation of the radial velocity component measurements present (mainly) in direction finding systems.

The HCSS and EACC variables are related to the parameters defining the variance and the accuracy of the radial velocity component measurements and they are mainly present in phased array systems. Thus, the two couples of variables are alternative, depending on the system producing the data.

RADAR_RADIAL & RADAR_TOTAL SEADATANET IDENTIFIERS

SeaDataNet (SDN) is the European project that federates the network of EU national oceanographic data centres. SDN is a data provider for Copernicus. Each SDN station distributed in Copernicus NetCDF data file have to include the following additional variables:

Table 6 – List of SeaDataNet additional variables

Name	Comment
SDN_CRUISE	Text string identifying the grouping label for the data object to which the data row belongs. For HFR data it is set equal to the site_code attribute, that is the EDIOS Series id of the HFR network.
SDN_STATION	Text string identifying the data object to which the data row belongs.

	For HFR data it is set equal to the platform_code attribute.
SDN_LOCAL_CDI_ID	The local identifier of the Common Data Index record associated with the data row.
SDN_EDMO_CODE	The key identifying the organization responsible for assigning the local CDI given in the European Directory of Marine Organizations (EDMO).
SDN_REFERENCES	Link to a single landing page - an XHTML document providing additional information.
SDN_XLINK	Text strings containing a URI (URN or URL) pointing to a web resource such as a usage metadata document for the data object to which the array element belongs.

RADAR_RADIAL & RADAR_TOTAL SDN NAMESPACE VARIABLES ATTRIBUTES

Variable attributes required in the SDC CF extension are part of the European common data and metadata model for NRT HFR current data. In particular, the SDN extensions to CF were concerned with providing storage for standardized semantics and metadata included in the SDN profiles format. The standardized semantics are included as four mandatory parameter attributes for each data or coordinate variable, which are:

Table 7- SeaDataNet mandatory variables attributes

Name	Comment
sdn_parameter_urn	URN (URL) for the parameter description taken from the P01 vocabulary.
sdn_parameter_name	Plain language label (Entryterm) for the parameter taken from the P01 vocabulary at the time of data file creation.
sdn_uom_urn	URN (URL) for the parameter units of measure taken from the P06 vocabulary.
sdn_uom_name	Plain language label (Entryterm) for the parameters' units of measure, taken from the P06 vocabulary at the time of data file creation.

According to SDC CF extension, the **ancillary_variables** attribute is mandatory and has to be set as the list of QC variables related to the specific variable.

ADCP VARIABLES

Table 8 – List of ADCP variables description

Variable name	Description	Units
TIME	Date of the data	days since 1950-01-01T00:00:00Z
DEPH	Depth of the temperature and sea water velocity (indicated in the file)	m
LATITUDE	Latitude of the data position	Degree North
LONGITUDE	Longitude of the data position	Degree East
EWCT	Eastward_sea_water_velocity	m s-1
NSCT	Northward_sea_water_velocity	m s-1
VCSP	Upward_sea_water_velocity	m s-1
TRAJECTORY	Trajectory, coordinate variable containing the cf_role	Dimensionless

ARGO VARIABLES

Table 9 - List of the available variables for Argo current dataset

Variable name	Description	Units
TIME	Date of the data	days since 1950-01-01T00:00:00Z
PRES	Representative pressure of the current observation	decibar
LATITUDE	Latitude of the data position	Degree North
LONGITUDE	Longitude of the data position	Degree East
EWCT	West-East sea water velocity at the representative pressure	m.s-1
NSCT	South-North sea water velocity at the representative pressure	m.s-1
TIME_INTERVAL	Time interval of the current variable (measured between two locations)	day
PLATFORM_CODE	The float platform code	n/a

GROUNDED	*Indicates the best estimate of whether the float touched the ground for that cycle. The conventions are described in Argo reference* Table 10.	n/a
TRAJECTORY	Trajectory, coordinate variable containing the cf_role	Dimensionless

Table 10- Argo reference ; GROUNDED flags

flag	Meaning
Y	Yes, the float touched the ground
B	Yes, the float touched the ground after bathymetry check with an outside database
N	No, the float did not touch the ground
S	Float is known to be drifting at a shallower depth than originally programmed
U	Unknown

GLIDER VARIABLES

Table 11 - List of the available variables for glider current dataset

Variable name	Description	Units
TIME	Date of the data	days since 1950-01-01T00:00:00Z
DEPH	Depth of the sea water velocity	m
LATITUDE	Latitude of the data position	Degree North
LONGITUDE	Longitude of the data position	Degree East

EWCT	West-East sea water velocity at the representative pressure	m.s-1
NSCT	South-North sea water velocity at the representative pressure	m.s-1
TRAJECTORY	Trajectory, coordinate variable containing the cf_role	Dimensionless

c. Content: quality control variables

QC variables (TIME_QC, POSITION_QC and DEPH_QC) for coordinate variables are mandated by the Copernicus Marine Service In Situ TAC profile.

No CF-1.11 standard names exist for QC variables; thus, long names are used. QC variables are of type byte.

QC variables (as defined in Table 12) can also exist not linked to a target physical variable (e.g. GDOP threshold QC variable linked to GDOP variable), but also as standalone variables reporting the results of a specific QC test (as detailed in the QUID report of this product), e.g. Over-water test.

Table 12- QC variables of the datasets included in this product.

Variable name	Long_name	Variable dimensionality	datasets					
			drifter	radar_radial	radar_total	adcp	argo	glider
TIME_QC	Time Quality Flag	scalar	x	x	x	x	x	x
POSITION_QC	Position Quality Flags	scalar or gridded	x scalar	x gridded	x gridded	x scalar	x	x
DEPH_QC	Depth Quality Flag	scalar	x	x	x	x		x
PRES_QC	Pressure Quality Flag	scalar					x	
EWCT_QC	West-east current component quality flag	scalar	x			x	x	x
NSCT_QC	South-north current component quality flag	scalar	x			x	x	x
VCSP_QC	Bottom-top current component quality flag					x		
EWCT_WS_QC	West-east current wind slippage correction at the drog depth quality	scalar	x					

	flag (=1: Computed data; = 9: No data computed)							
NSCT_WS_QC	South-north current wind slippage correction at the drog depth quality flag (=1: Computed data; = 9: No data computed)	scalar	x					
QCflag	Overall Quality Flags	gridded		x	x			
VART_QC	Variance Threshold Quality Flags	gridded		x	x			
CSPD_QC	Velocity Threshold Quality Flags	gridded		x	x			
GDOP_QC	GDOP Threshold Quality Flags	gridded			x			
DDNS_QC	Data Density Threshold Quality Flags	gridded			x			

QUALITY CONTROL FLAGS

The quality control flags are common for the six datasets included in the product and they indicate the quality of the data values in a file and are assigned after quality control procedures have been performed (as detailed in the QUID manual of the current product).

When QC information is provided as a separate flag variable, CF-1.11 requires that these variables carry the "flag_values" and "flag_meanings" attributes. These flags or codes are used in the <DATA>_QC variables to describe the quality of each measurement. The QC flag scale and definition is detailed in *Copernicus Marine In Situ TAC quality control procedure (https://doi.org/10.13155/106219).*

Some specifies has to be considered for some datasets:

- For **drifter:**
  - the wind-slippage correction of the velocity data (=1: computed data; = 9: no data computed) is provided by the NSCT_WS_QC and EWCT_WS_QC.
  - DEPH_QC does not follow any QC test, it is indicative of the reliability of the depth of the data, which depends on the DROGUE_TEST: DEPH_QC =1 when DROGUE_TEST = 1 or 3; DEPH_QC =3 when DROGUE_TEST = 0.
- For **radar_radial** & **radar_total**:
  - A file with no valid time, depth and position is not distributed on Copernicus In Situ TAC.
  - A valid time, depth and position have a TIME_QC, DEPH_QC and POSITION_QC variables, respectively, set to 1, 2, 5, 7 or 8 (good, probably good, value changed, nominal, interpolated).
  - Since the datasets radar_total are gridded data, they are distributed even if the Qcflag (overall QC flag for the file) is not entirely filled with good_data (qc_flag=1). This is because it may happen that some vectors on the grid are labelled as good_data and some other as bad_data. Users can apply the QC variables (there is one QC variable per each QC test applied to the data) as masks for the geophysical variables.
- For **adcp:**
  - Dates and positions are flagged good (QC = 1) or bad (QC = 4), and only these 2 flags values are used.
  - Same for current variables eastward_sea_water_velocity, northward_sea_water_velocity, upward_sea_water_velocity and sea_water_depth
- For **argo**:

- The data distributed have positions and dates quality flags equal to 1 (good data). The initial argo data used to compute the velocities with positions and dates that do not have a QC 1 are ignored

- For **glider**:
  - Dates and positions are flagged good (QC = 1) or bad (QC = 4), and only these 2 flags values are used.

d. Content: global attributes

The global attribute section of a NetCDF file describes the contents of the file overall and allows for data discovery. All fields should be human-readable and use units that are easy to understand. Global attribute names are case sensitive.

Attributes are organized by function: Discovery and Identification, Geo-spatial-temporal, Conventions used, Publication information, and Provenance.

These global attributes are described in the [DR3] (see sect. 5)).

SPECIFIC GLOBAL ATTRIBUTES

The datasets contain common attributes and specific attributes.

Very specific global attributes comprehend those present in only certain datasets (either drifter, adcp, radar_total or radar_radial, glider, argo) and listed in the dedicated tables. They are also documented in the [DR3] (see sect. 5)). These specific attributes are highlighted in this section.

SPECIFIC DRIFTER DATASET GLOBAL ATTRIBUTES

Table 13 - Specific global attributes for the drifter dataset

Geo-spatial-temporal	
Attribute	***Description***
start_mission_date	Date of the platform deployment, if available
End_mission_date	Date of the end of period the drifter was freely drifting in the ocean, if available
estimated_drogue_off_date	Date of the platform drogue lost, if any
Transmitter_system	Transmitter system of the locations (Argos or Iridium)

SPECIFIC RADAR_RADIAL & RADAR_TOTAL DATASETS GLOBAL ATTRIBUTES

Table 14- Specific global attributes for the radar_radial and radial_total datasets

Discovery and Identification	
Attribute	***Description***
doa_estimation_method	Specifies if the system is Direction Finding or Beam Forming. Possible values are “Direction Finding” and “Beam Forming”.
calibration_type	Specifies if calibration has been performed. Possible values are: “None”, “Ideal”, “APM”, “full”, “internal”, “physical”, “AEA”.
last_calibration_date	Reports the date of the last calibration. It must be specified as a string in the ISO8601 standard "YYYY-MM-DD-Thh:mm:ssZ". UTC must be used, and specified.
calibration_link	Indicates the link to a contact person able to provide data about the calibration.
network	A grouping of sites based on common shore-based logistics or infrastructure.
Geo-spatial-temporal	
Attribute	***Description***
geospatial_vertical_resolution	Vertical resolution of the measurement. For HFR data it is set as the maximum integration depth of the radar measurement, according to its operating frequency.
Provenance	
Attribute	***Description***
processing_level	Level of processing and quality control applied to data. The radar_total data set delivered corresponds to LEVEL 3B data, following the definition of the processing levels for the identification of the different data produced during the processing workflow of a HF radar. Level 3B are surface currents mapped on uniform space-time grid scales and that have been processed with a minimum set of QC.[4]

[4] Please refer to Appendix A of the JERICO-NEXT deliverable D5.14 “Recommendation Report 2 on improved common procedures for HFR QC analysis” for the processing level definition (as summarized in Table 10; http://www.jerico-ri.eu/download/jerico-next-deliverables/JERICO-NEXT-Deliverable_5.14_V1.pdf)

SPECIFIC ADCP DATASET GLOBAL ATTRIBUTES

There are no specific global attributes for adcp.

SPECIFIC ARGO DATASET GLOBAL ATTRIBUTES

There are no specific global attributes for argo.

SPECIFIC GLIDER DATASET GLOBAL ATTRIBUTES

There are no specific global attributes for glider.

d) Data distribution

a. Data organization

The Global Delayed Mode In-situ Surface currents and velocity profiles dataset is available by accessing the **INSITU_GLO_PHY_UV_DISCRETE_MY_013_044** product directory.

Data coming from drifters, HFR, vessel mounted ADCP platforms, argo floats and gliders are grouped in 6 different datasets (Figure 1):

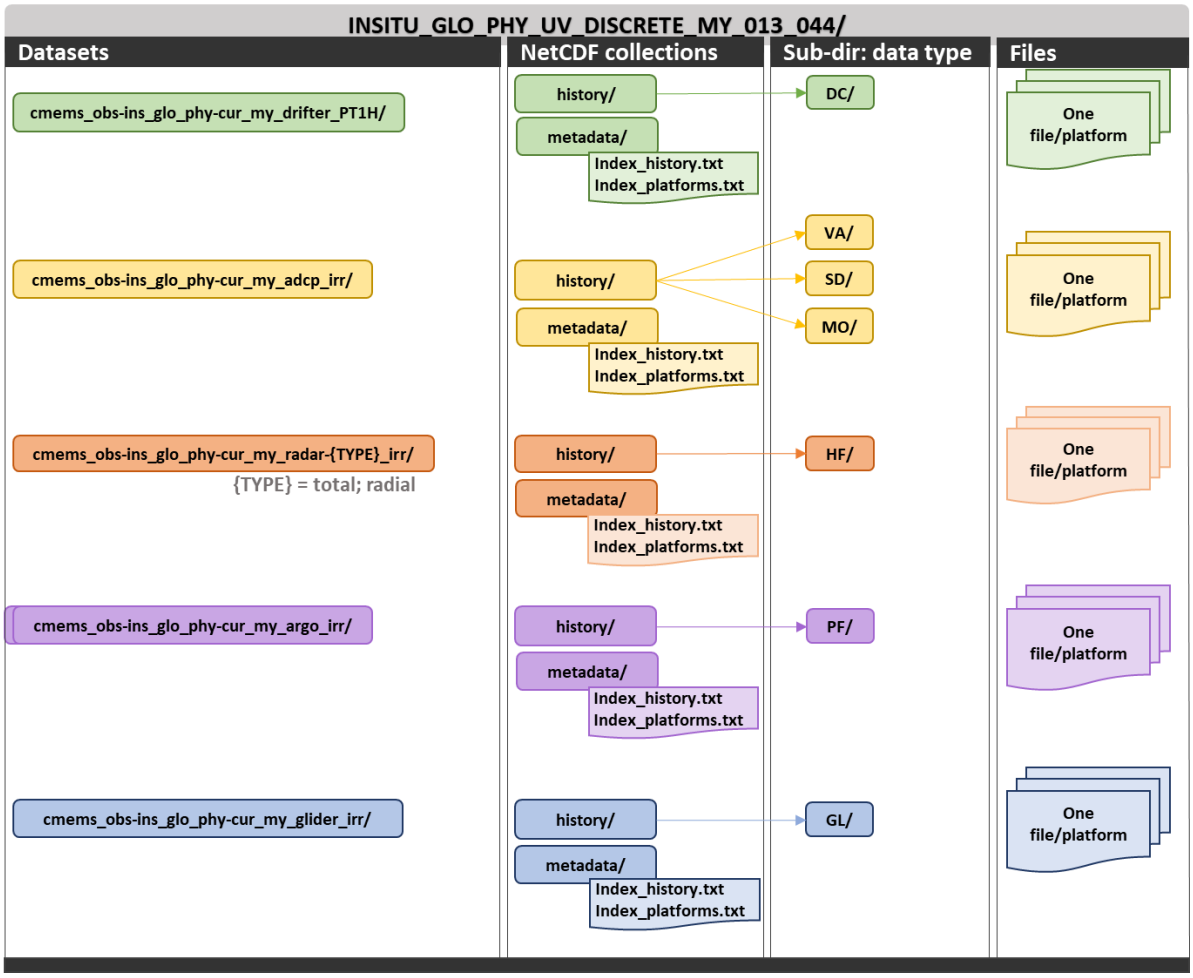

Figure 1: Directory tree of INSITU_GLO_PHY_UV_DISCRETE_MY_013_044 product. See sect. 3) for data_type definition

b. Index files

It is important to help users find the platforms and the files that provide such observations in the dataset. A data discovery mechanism is provided by means of index files, which are located at the top level of the dataset directory and provide information on each platform and file located on the server.

These index files allow users to know what is in the files without having to download them. They contain a list of all available data files by providing a relative path to the file, together with a selection of relevant metadata (temporal and spatial ranges, parameters…) that describes the data file contents.

One index_history.txt file per dataset describe the content of history directories.

In addition, another index file (index_platform.txt) is provided. It registers the list of all the individual platforms that are available in the product directory.

These index files are also useful for automatic data download by operational users.

INDEX FILES UPDATE AND CONSISTENCY

There is consistency between the index files and the file system. When a file must be deleted, its reference is removed from the index before the file deletion. When a file is added or updated, it is indexed after its addition or update.

DESCRIPTION OF THE INDEX HISTORY FILE CONTENT

Each index_history.txt file contains one line per file exposing the *file_name* for downloading plus some metadata for early filtering. The lines in the index are sorted by file name and time coverage start. These files are updated at each new release of the datasets.

- product_id: COP-XX-YY
  - COP: Copernicus trigram
  - XX: region bigram (GL)
  - YY: product version

Example: COP-GL-02

- file_name: relative path of the file to access the data, with the format: <product_name>/<dataset_name>_<release_version_tag>/history/<file_name>
- geospatial_lat_min
- geospatial_lat_max
- geospatial_lon_min

- geospatial_lon_max
- time_coverage_start
- time_coverage_end
- institution
- date_update
- data_mode:
    - D: delayed-mode data. Data published after all calibrations and quality control procedures have been applied on the internally recorded or best available original data. This is the best possible version of processed data.
- Parameters (separator: blank)

ISO8601 format is used in date-time fields: YYYY-MM-DDThh:mm:ssZ

The **header** is composed of several lines starting with '#' character. It contains metadata about the content of the index file.

Title : in-situ files catalog

Description : catalog of available in-situ files

Project : Copernicus Marine In Situ TAC

Format version : 1.4

Date of update : 2020-04-20T12:34:20Z

#catalog_id,file_name,geospatial_lat_min,geospatial_lat_max,geospatial_lon_min, geospatial_lon_max,time_coverage_start,time_coverage_end,institution,date_update,data_ mode, parameters

Copernicus Marine Service In Situ data file index_history.txt example (for drifter)
Title : in-situ files catalog # # Description : catalog of available in-situ files. # # Project : Copernicus Marine In Situ TAC # # Format version : 2.0 # # Date of update : 2023-09-26T21:14:41Z # #catalog_id,file_name,geospatial_lat_min,geospatial_lat_max,geospatial_lon_min,geospatial_lon_max, time_coverage_start,time_coverage_end,institution,date_update,data_mode,parameters

```
COP-GL-02,INSITU_GLO_PHY_UV_DISCRETE_MY_013_044/cmems_obs-ins_glo_phy-
cur_my_drifter_PT1H_202311/history/GL_TS_DC_99010_4600588.nc,43.215,45.455,189.996,212.031,2
10-12-22T00:00:00Z,2011-08-31T12:00:00Z,AOML,2019-10-18T09:44:53Z,D, LONGITUDE LATITUDE
TIME TIME_QC DEPH DEPH_QC POSITION_QC EWCT EWCT_QC NSCT NSCT_QC EWCT_WS
EWCT_WS_QC NSCT_WS NSCT_WS_QC WS_TYPE_OF_PROCESSING TEMP TEMP_QC,......
```

DESCRIPTION OF INDEX PLATFORM INDEX FILE

The platforms index files are updated at each release of the datasets, and they register the list of individual platforms that are available on the server.

Each platform index contains a line per platform with the following information:

- platform_code
- date_creation: date of the first file related to the platform appearing in the product
- date_update: the latest update date of any file associated with the platform
- wmo_platform_code: It can be empty.
- data_source: list of different data streams of the platform = all possible combinations of data types, file type and sources (latest, monthly, history) of this platform (see the example of a line content below)
- institution
- institution_edmo_code: list of the different edmo codes of the platform. It can be empty.
- parameters
- last_latitude_observation
- last_longitude_observation
- last_date_observation

The information in both index files and NetCDF files must be the same when the index element is a global attribute in the NetCDF file. It implies that the computation of these global attributes is performed in the generation process of the NetCDF files.

A platform will appear in the platforms index only if there is at least one file in the files' indexes.

It must be cross-checked that platform_code used in files indexes matches one of the platfom_code listed in the index_platform.txt.

The **fields** are provided **in the order** mentioned in this definition.

The **index lines are sorted** by platform_code.

The **field separator** character is "," (comma). Within each field, the commas found in the content are replaced by "-". Some fields contain a list of values separated by blank (example: the list of parameters) or separated by semicolon (if blank is a valid character in the values).

Data streams are identified thanks to the different filenames without date/period value.

ISO8601 format used in date-time fields: YYYY-MM-DDThh:mm:ssZ

The **header** is composed of several lines starting with '#' character. It contains metadata about the content of the index file.

Title : in-situ platforms catalog

Description : catalog of available in-situ platforms

Project : Copernicus Marine In Situ TAC

Format version : 1.4

Date of update : 2020-04-20T12:36:44Z

platform_code,date_creation,date_update,wmo_platform_code,data_source,institution, institution_edmo_code,parameters,last_latitude_observation,last_longitude_observation, last_date_observation

Copernicus In Situ data platform index example (for drifter)
Title : in-situ platforms catalog # Description : catalog of available in-situ platforms # Project : Copernicus Marine In Situ TAC # Format version : 1.4 # Date of update : 2023-11-08T15:53:11Z # platform_code,date_creation,date_update,wmo_platform_code,data_source,institution,institution_edmo_code,parameters,last_latitude_observation,last_longitude_observation,last_date_observation 10033430,,2023-09-26T18:31:41Z,3201874,GL_TS_DC_10033430_3201874,AOML,1799,DEPH EWCT NSCT EWCT_WS NSCT_WS TEMP,-7.37,-156.75,2023-06-15T16:59:00Z 10034430,,2023-09-26T18:37:46Z,5102831,GL_TS_DC_10034430_5102831,AOML,1799,DEPH EWCT NSCT EWCT_WS NSCT_WS TEMP,23.2,172.74,2023-06-15T16:59:00Z 10037420,,2023-09-26T18:38:41Z,5102832,GL_TS_DC_10037420_5102832,AOML,1799,DEPH EWCT NSCT EWCT_WS NSCT_WS TEMP,-2.7,152.01999999999998,2023-05-11T19:59:00Z 10050130,,2023-09-26T18:41:24Z,5600935,GL_TS_DC_10050130_5600935,AOML,1799,DEPH EWCT NSCT EWCT_WS NSCT_WS TEMP,-23.15,108.06,2012-07-14T18:00:00Z 10051120,,2023-09-26T18:41:44Z,5600524,GL_TS_DC_10051120_5600524,AOML,1799,DEPH EWCT NSCT EWCT_WS NSCT_WS TEMP,-56.66,138.51999999999998,20

e) Catalogue updating

This product is updated twice a year, with time series extension, additional observations or full-reprocessing of the entire database.

f) Product use and impact

This product is designed to be assimilated into or for validation purposes of operational models operated by ocean forecasting centers, for reanalysis purposes or for research community. These users need data aggregated and quality controlled in a reliable and documented manner.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

The name convention of the six different datasets included in the product is as following:

Directory	Naming convention = RR_XX_YY_CODE.nc	Meaning[5]
cmems_obs-ins_glo_phy-cur_my_drifter_PT1H	E.g.: GL_TS_DC_99450.nc	• **RR:** Region Bigram, • **XX:** File type • **YY:** Data type source • **CODE: unique** Platform code identifier • .nc: NetCDF file name suffix
cmems_obs-ins_glo_phy-cur_my_radar-radial_irr	E.g.: GL_RV_HF_HFR-EUSKOOS-HIGE.nc	
cmems_obs-ins_glo_phy-cur_my_radar-total_irr	E.g.: GL_TV_HF_HFR-TirLig.nc	
cmems_obs-ins_glo_phy-cur_my_adcp_irr	E.g.: GL_TS_VA_FMCY.nc	
cmems_obs-ins_glo_phy-cur_my_argo_irr	E.g.: GL_TS_PF_7901091.nc	
cmems_obs-ins_glo_phy-cur_my_glider_irr	E.g.: GL_TS_GL_6801586.nc	

- **Region bigrams (RR)**

[5] From the Copernicus In Situ TAC SRD - System Requirement Document: http://dx.doi.org/10.13155/40846

  - GL: global Production Unit (PU) of the In Situ TAC.
- **File types bigrams (XX)**
  - TS: Time Series or Trajectories
  - TV: gridded total velocities
  - RV: gridded radials velocities
- **Data types bigrams (YY)**
  - DC: drifter buoy reporting calculated sea water current
  - HF: high frequency radar
  - VA: vessel-mounted ADCP
  - PF: Argo floats
  - SD: Saildrones ADCP
  - MO: mooring ADCP
  - GL : Gliders
- **CODE**
  - The platform code or ID is a unique identifier of the platform within all Copernicus Marine Service In Situ TAC production units. For other identifiers the user should dive into the global attributes of the file where other codes are exposed if available. For drifters, this platform code is the WMO code. If the WMO code is a re-used number for the drifter, the In Situ TAC adds an additional letter (A, B, C,…) in order to have a unique identifier (ex: GL_TS_DC_1300629A.nc)

The file naming convention is very useful for finding a particular file from a drifter's trajectory, an HFR system or a vessel-mounted adcp survey.

- In the directory of the **cmems_obs-ins_glo_phy-cur_my_drifter_PT1H dataset**, each file corresponds to a particular drifter trajectory, as indicated by the platform ID included in the file name.
- In the directory of the **cmems_obs-ins_glo_phy-cur_my_radar-radial_irr dataset**, each file corresponds to a particular HFR station from a particular HFR network, as indicated by the site ID included in the filename.
- In the directory of the **cmems_obs-ins_glo_phy-cur_my_radar-total_irr dataset**, each file corresponds to a particular HFR system, as indicated by the platform ID included in the filename.

- In the directory of the **cmems_obs-ins_glo_phy-cur_my_adcp_irr dataset**, each file corresponds to a particular platform. ADCP, Moorings and Saildrones data type are available. The file name contains the WMO code of the platform.
- In the directory of the **cmems_obs-ins_glo_phy-cur_my_argo_irr dataset**, each file corresponds to a particular Argo trajectory, as indicated by the platform ID included in the filename.
- In the directory of the **cmems_obs-ins_glo_phy-cur_my_glider_irr dataset**, each file corresponds to a particular glider trajectory, as indicated by the platform ID included in the filename.

5) REFERENCES

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[2] Copernicus Marine in situ TAC - **Physical Parameters List** https://doi.org/10.13155/53381

[3] Copernicus Marine in situ **NetCDF Attributes list** https://doi.org/10.13155/95044

[4] Copernicus in situ TAC - **CMEMS regions definition** https://doi.org/10.17882/44395

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